

Engineering Reboot 2026

Day 23 Engineering Workbook

Version 1.1 – Implementation Edition (Locked)

Mission

Organize validated data using Python collections and build an Employee Directory application that evolves through Week 5.

Learning Objectives

Explain lists, tuples, sets and dictionaries.

Choose the right collection.

Iterate, search, update and display structured data.

Combine validation with collection management.

Prepare for Days 24 and 25.

Week 5 Engineering Scenario

Maintain an Employee Directory application using one canonical dataset throughout Days 23–25.

Week 5 Canonical Dataset

```
EMPLOYEES = [{'id':1001,'name':'Alice Johnson','age':28,'department':'Engineering','salary':85000}, {'id':1002,'name':'Bob Smith','age':34,'department':'Finance','salary':72000}, {'id':1003,'name':'Charlie Brown','age':41,'department':'Human Resources','salary':69000}, {'id':1004,'name':'Diana Miller','age':30,'department':'Marketing','salary':78000}, {'id':1005,'name':'Ethan Davis','age':37,'department':'Engineering','salary':92000}]
```

Exercises

Exercise 1: Manipulate the EMPLOYEES list.

Exercise 2: Build a department set from EMPLOYEES.

Exercise 3: Update an employee dictionary.

Mini Project

Build an Employee Directory supporting display, search by ID/name, department filtering and employee count.

Stretch Goals

Average salary

Highest-paid employee

Sort alphabetically

Sort by salary

Prevent duplicates

Git Commit

Day 23 - Completed collection fundamentals and employee directory project